Weiyang Jin

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EDUCATION

The University of Hong Kong, Hong Kong SAR, China **2025.11 – Present**
Ph.D. Student, IDS (HKU-MMLab) (Advisor: Prof. Xihui Liu)

RESEARCH INTERESTS

World models, multimodal generation, and visual intelligence: how to build visual intelligence via representation learning and self-evolution.

PUBLICATIONS AND PREPRINTS

* Equal contribution † Project leader ‡ Corresponding author

[1] **SRUM: Fine-Grained Self-Rewarding for Unified Multimodal Models** *ECCV 2026* [Paper] [Code]
Weiyang Jin*, Yuwei Niu*, Jiaqi Liao, Chengqi Duan, Aoxue Li, Shenghua Gao, Xihui Liu
- Self-rewarding post-training where a unified model's own understanding module rewards its generation through global and local rewards; T2I-CompBench **82.18** → **88.37**.

[2] **UniVerse: Empower Unified Generation with Reasoning and Knowledge** *CVPR 2026* [Paper] [Code]
Kaiyue Sun*, **Weiyang Jin***, +6 others, Xihui Liu
- Equips unified multimodal models with explicit reasoning and world knowledge, so generation follows what the model understands rather than surface prompt matching.

[3] **Does Understanding Inform Generation in Unified Multimodal Models? From Analysis to Path Forward** *EMNLP 2026 Main* [Paper] [Code]
Yuwei Niu*, **Weiyang Jin***, +7 others, Li Yuan
- A decoupled sandbox with leakage-free synthetic data that exposes the understanding–generation gap; chain-of-thought and self-training close it for reasoning and knowledge transfer.

[4] **WISE: A World Knowledge-Informed Semantic Evaluation for Text-to-Image Generation** *ICML 2026* [Paper] [Code]
Yuwei Niu*, Munan Ning*, Mengren Zheng, **Weiyang Jin***, +7 others, Li Yuan
- First world-knowledge benchmark for text-to-image generation: **1,000** prompts over **25** subdomains with the WiScore metric, evaluating **20** T2I and unified models.

[5] **Execution Before Memory: Diagnosing Memory Failures in Video World Models** *Under Review* [Paper] [Code]
Weiyang Jin, Jiwen Yu, Sihan Xu, Yuwei Niu, Xihui Liu
- Separates instruction execution from memory in video world models and shows execution is the bottleneck; M3-Agent more than **triples** the MIST reveal score.

[6] **Echo-Memory: A Controlled Study of Memory in Action World Models** *Under Review* [Paper] [Code]
Wayne King*, +14 others, Nan Duan*
- Matched comparison of memory designs for action world models under one backbone, separating capacity, compression, read-out, and recurrence across replay and return probes.

[7] **EchoWM: Open and Enterable Omnimodal World Models** *Under Review* [Paper] [Code]
Songchun Zhang*, Yaowei Li*, Junhao Zhuang*, **Weiyang Jin***, +15 others, Zeyue Xue*, Anyi Rao†, Nan Duan†
- Enterable omnimodal world model that jointly generates **720p** video, environmental sound, music, and speech under continuous 6-DoF navigation.

[8] **Next-Embedding Prediction Makes Strong Vision Learners** *Under Review* [Paper] [Code]
Sihan Xu, Ziqiao Ma, Wenhao Chai, Xuweiyi Chen, **Weiyang Jin**, Joyce Chai, Saining Xie, Stella X. Yu
- Generative visual pretraining by next-embedding prediction alone, without pixels, tokens, or contrastive loss; **83.8%** / **85.3%** ImageNet-1K top-1 with ViT-B / ViT-L.

TECHNICAL REPORTS

[9] **OpenWAM** *Technical Report · CoRL 2026 TAE Best Paper Award* [Paper] [Code]
- Open, modular stack that turns world–action pretraining into controlled experiments; OpenWAM-α is pretrained on about **6,400** hours of egocentric human and robot data.

[10] **InternVLA-M1** ***Technical Report*** [Paper] [Code]
- Spatially guided vision-language-action training from **2.3M** spatial reasoning samples; +**14.6%** on SimplerEnv Google Robot and +**20.6%** on unseen real-world objects.

[11] **RoboDojo** ***Technical Report*** [Paper] [Code]
- Unified sim-and-real benchmark for generalist manipulation policies: **42** simulation and **18** real-world tasks, with **30** policies on a public leaderboard.

[12] **JoyAI-Series** ***Technical Report*** [JoyAI-Echo] [JoyAI-Echo-1.5] [JoyAI-Image-Edit]
- Long-horizon audio-visual generation with cross-shot memory (Echo, Echo-1.5) and a unified model for image understanding, generation, and editing (Image-Edit).

OPEN-SOURCE PROJECTS

Omni-Rewriter, Owner [Code]
Agentic prompt-expansion harness for image and video generation, from demos to deployable workflows.

LMMs-Eval, Contributor [Code]
One-for-all multimodal evaluation toolkit across text, image, video, and audio tasks.

RESEARCH EXPERIENCE

Joy Future Academy, Echo Team, Research Intern, advised by Dr. Haoyang Huang 2025.11 – Present
World models, long-horizon audio-visual generation, and memory in interactive rollouts (Top Genius Program).

Shanghai AI Laboratory, Intern Robotics, Research Intern, advised by Dr. Yilun Chen 2025.05 – 2025.08
Generalist robot policies and spatially guided VLA systems (InternVLA-M1).

New York University, Vision-X Lab, Research Intern, advised by Prof. Saining Xie 2024.02 – 2025.05
Visual representation learning and next-embedding predictive architectures (NEPA).

Westlake University, CAIRI Lab, Research Intern, advised by Prof. Stan Z. Li 2023.04 – 2024.02
Machine learning and visual intelligence.

Tsinghua University, AIR, Research Intern, advised by Prof. Hao Zhao 2022.05 – 2023.04
Computer vision and embodied intelligence.

INVITED TALKS

From Digital World to Physical Intelligence: Party After Frontier LLMs 2026.09
Amap (Gaode), Alibaba · Invited by Dr. Yilun Chen

Our Vision is Dead? 2025.10
Shenzhen Loop Area Institute (SLAI) · Invited by Prof. Xiaoguang Han

SERVICES

Cofounder, Lumina, an embodied intelligence open-source community

Conference Reviewer, NeurIPS, ICML, ICLR (2025–2026); CVPR, ICCV, ECCV (2024–2026)

Journal Reviewer, Pattern Recognition

BLOG

How Far Are We from a Native Autoregressive Video Model? 2026.08.31
A note on what native autoregression should mean for video, and how far current stacks still are.